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Q2. using formula out = $(W - K + 2P)/S + 1$

$= (32 - 5 + 2)/1 + 1$

$= 28$

so output is 28×28

Q1. In deep networks during backprop the gradient gets multiplied by derivative of activation at each layer.

~~sigmoid derivative is 1 so it is fine~~

For sigmoid the derivative is at most 0.25 so after many layers the gradient becomes very small and the first layers stop learning. This is vanishing gradient.

We can use ReLU activation to reduce it.

Q3.

```
def softmax(x):
    import math
    exps = [math.exp(v) for v in x]
    total = sum(exps)
    return [e/total for e in exps]
```